Customer Segmentation Analysis

(Postgraduate predictive analytics project, 2025)

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I. Introduction

In today's competitive fitness market, generic marketing often fails to engage diverse members' needs. This report segments 2,000 gym members using K-means++ and Agglomerative Clustering, based on demographics like age, income, education, and location. After standardising variables and validating results through the Elbow Method and Silhouette Analysis, two distinct segments emerged. These findings support targeted strategies: affordable plans for budget-conscious members and premium services for affluent professionals, helping the gym enhance engagement, retention, and revenue growth.

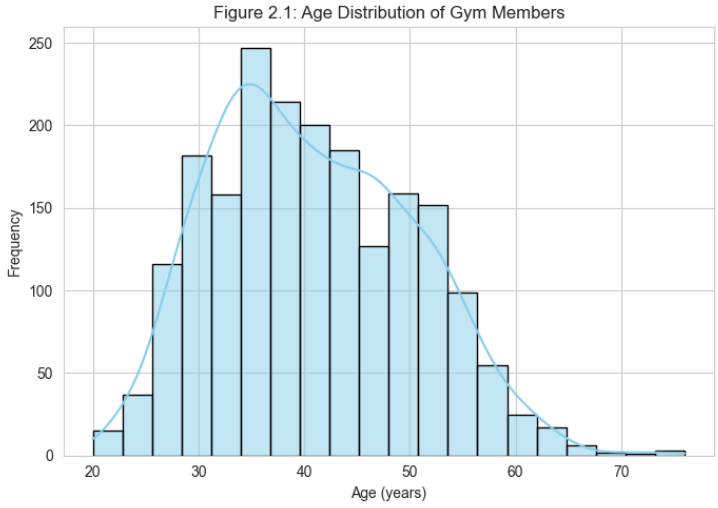
II. Exploratory Data Analysis

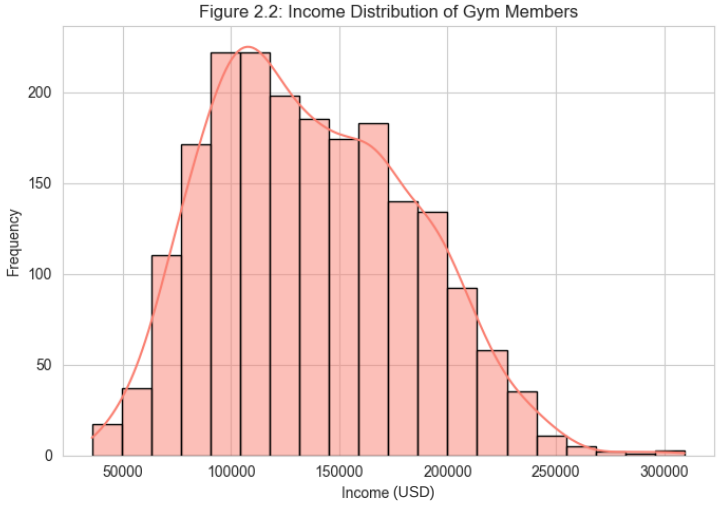
Table 2.1 shows that the average customer age is 40.82. The average income is $137,516, ranging between $35,832 and $309,364, indicating high variability.

	count	mean	std	min	25%	50%	75%	max
Age	2000.00	40.82	9.46	20.00	33.00	40.00	48.00	76.00
Income	2000.00	137516.20	46184.30	35832.00	101262.75	133004.00	171232.50	309364.00

Table 2.1: Summary Statistics for Numerical Variables

Figure 2.1 shows that the age distribution is slightly right-skewed, between 30 and 50 years, with a gradual decline beyond 60. Figure 2.2 displays a right-skewed income distribution, with a strong concentration around $100,000–$150,000 and a long tail stretching above $250,000, indicating a core middle-income group and a smaller high-income minority.





Variable	Category	Count	Percentage (%)
Gender	Female	1209	60.45
	Male	791	39.55
Marital Status	Non-single	1001	50.05
	Single	999	49.95
Education	Graduate school	174	8.70
	High school	877	43.85
	Other/Unknown	192	9.60
	University	757	37.85
Occupation	Management/Self-employed	217	10.85
	Skilled Employee/Official	791	39.55
	Unemployed/Unskilled	992	49.60
Settlement Size	Big city	798	39.90
	Mid-sized city	72	3.60
	Small city	1130	56.50

Table 2.2: Summary Statistics for Categorical Variables

As shown in Table 2.2, the overall demographics of the dataset show that most members are female, live in small cities, work in skilled occupations, and have either a high school or university education.

Figure 3: The Distribution of Attributes

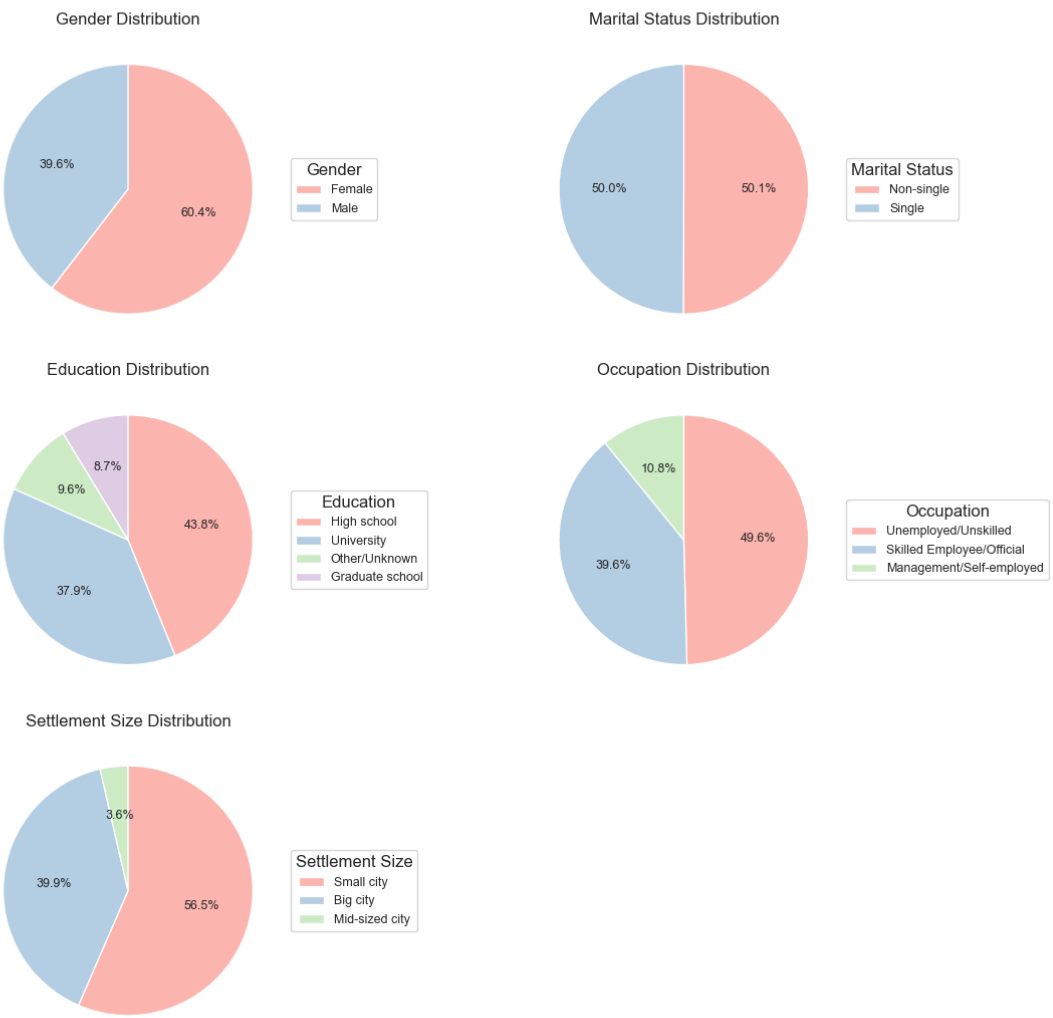
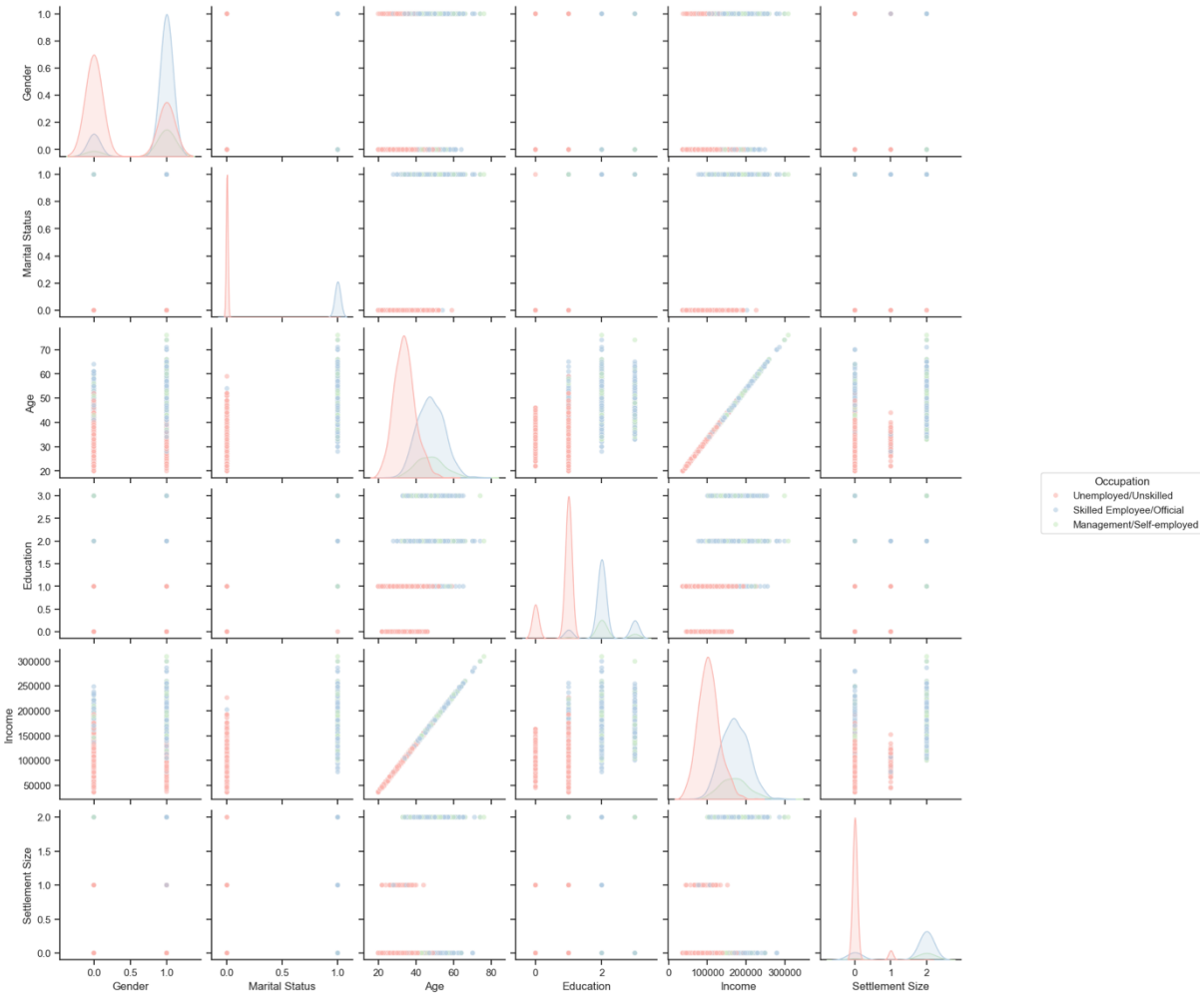


Figure 2.3 emphasises the demographic imbalances, particularly the higher proportion of females and non-single individuals and the concentration of members with high school or university education. It also highlights the underrepresentation of mid-sized cities among settlement types and shows a relatively balanced split between unskilled roles and those with skilled or managerial positions.

Figure 2.4: Customer Demographics by Occupation



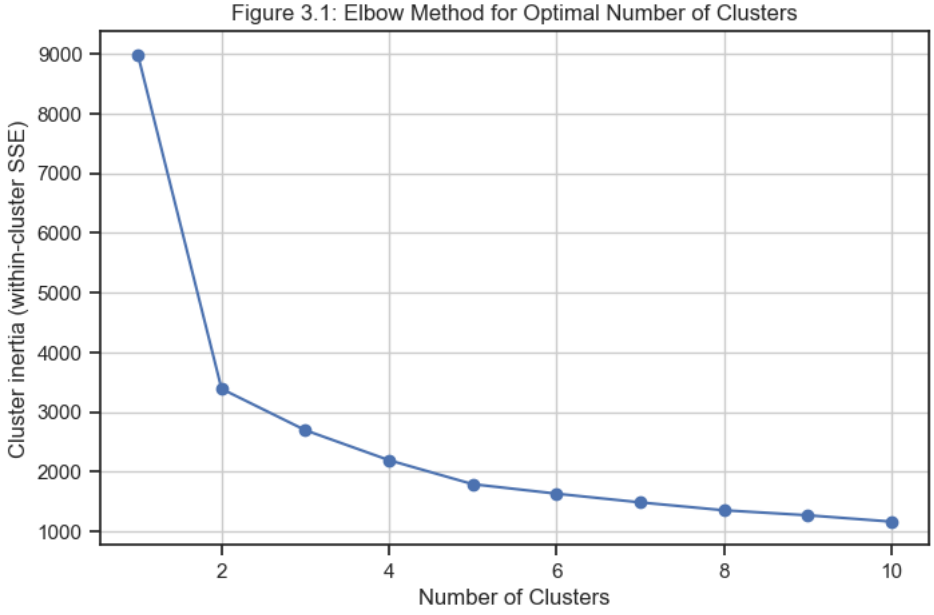
The pair plot, as shown in Figure 2.4, highlights that younger, less educated individuals with lower income and small-city residence tend to be in unskilled roles. In contrast, skilled and managerial members are generally older, better educated, earn higher incomes, and are more concentrated in large cities. Age and income show a positive linear relationship; education is closely linked to occupation and income. These patterns support clustering by showing how key features interact to form distinct customer groups.

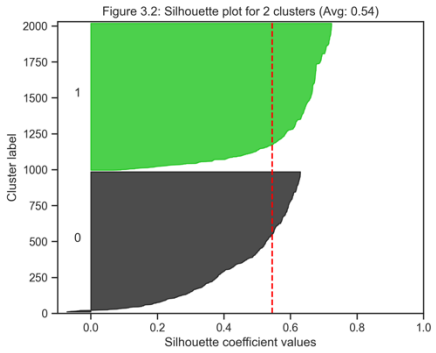
III. Customer Segmentation

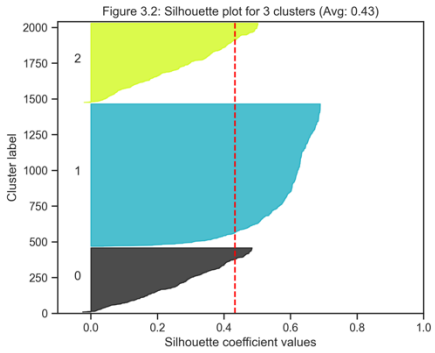
The numerical variables Age and Income were standardised using StandardScaler to prepare the dataset for clustering. This step was essential to ensure both features contributed equally to the clustering process, especially given Income's larger scale.

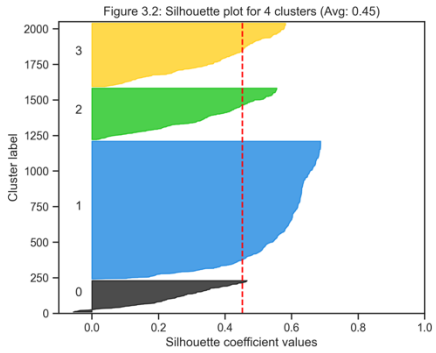
Both the Elbow Method and Silhouette Analysis were used to determine the optimal number of customer segments. The Elbow plot (Figure 3.1) shows a sharp decrease in within-cluster variation when moving from one to two clusters. This suggests that two clusters capture the main structure in the data. Supporting this, the Silhouette Analysis (Figure 3.2 – 2 clusters) shows the highest average silhouette score of 0.54 at k = 2, indicating well-separated and cohesive clusters. Although k=3 and k = 4 were also tested, their lower silhouette scores (0.43 and 0.45, respectively) suggested less distinct segments. Based on this, two clusters were selected as the optimal and most interpretable solution.

Figure 3.1: Elbow Method for Optimal Number of Clusters









K-means++ and Agglomerative Clustering were then applied to the standardised dataset using all variables without dimensionality reduction. The results were summarised below:

3.1. K-means++

	KMeans Cluster 0	KMeans Cluster 1	Interpret KMeans Cluster 0	Interpret KMeans Cluster 1
Count	1024.00	976.00	1024	976
Gender	0.37	0.86	Male	Female
Marital Status	0.03	0.99	Single	Non-single
Age	33.81	48.18	33.81	48.18
Education	0.84	2.10	High school	University
Income	103256.60	173460.69	103256.60	173460.69
Occupation	0.04	1.21	Unemployed/Unskilled	Skilled Employee/Official
Settlement Size	0.07	1.64	Small city	Big city

Table 3.1: Demographic Summary of Customer Segments (K-Means Clustering, k = 2)

3.2. Agglomerative Clustering

	Agglomerative Cluster 0	Agglomerative Cluster 1	Interpret Agglomerative Cluster 0	Interpret Agglomerative Cluster 1
Gender	0.86	0.35	Female	Male
Marital Status	0.99	0.00	Non-single	Single
Age	47.70	33.83	47.70	33.83
Education	2.10	0.81	University	High school
Income	171149.22	103340.70	171149.22	103340.70
Occupation	1.21	0.00	Skilled Employee/Official	Unemployed/Unskilled
Settlement Size	1.59	0.06	Big city	Small city
Count	1008.00	992.00	1008	992

Table 3.2: Demographic Summary of Customer Segments (Agglomerative Clustering, k = 2)

3.3. Segment Interpretation

Clustering Method	Cluster	Segment Name	Key Characteristics
K-means++ Clustering	0	Budget-Conscious Starters	Young, single males; high school education; modest income; mostly unemployed or unskilled; predominantly live in small cities.
	1	Established Urban Professionals	Middle-aged, non-single females; university-educated; high income; skilled professionals; live in big cities
Agglomerative Clustering	0	Established Urban Professionals	Mirrors K-means Cluster 1; older, educated, high-income women in big cities with skilled occupations.
	1	Budget-Conscious Starters	Matches K-means Cluster 0; younger, lower-income males with lower education, residing in small towns and often unemployed.

Table 3.3: Segment Interpretation Summary

3.4. Methods Comparison

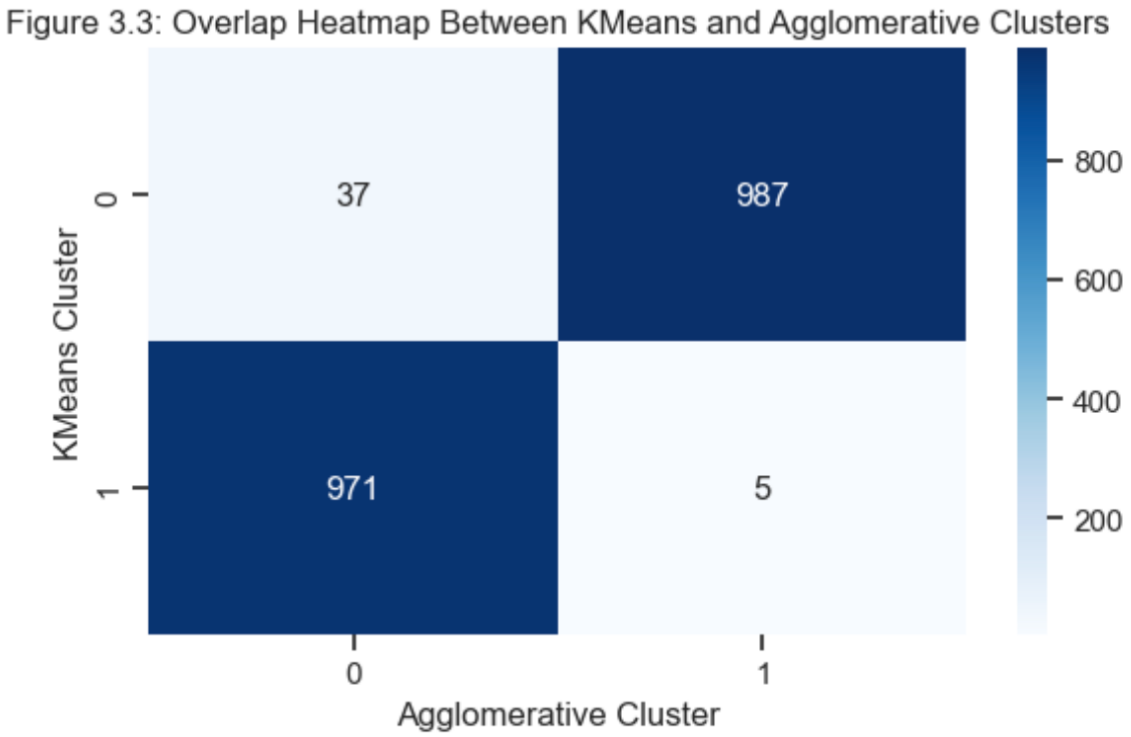


Figure 3.3 illustrates a near-perfect overlap (97%) between the two clustering methods, confirming the robustness of the segmentation. Most members in KMeans Cluster 0 align with Agglomerative Cluster 1, and KMeans Cluster 1 aligns with Agglomerative Cluster 0. Both methods consistently identify two distinct segments: younger, single males with lower income and education levels living in small towns, and older, non-single females with higher income, education, and urban residence.

IV. Market Strategies

For **Budget-Conscious Starters**, the gym chain should prioritise affordability and community engagement to resonate with this demographic's financial constraints and social dynamics. Introducing a tiered pricing model with verified student or part-time worker discounts would address their limited disposable income, while pay-as-you-go plans or short-term 3-month contracts could reduce upfront financial barriers. To foster loyalty, the gym could partner with local small-town businesses—cafes, sports retailers, or cinemas—to offer exclusive member perks, like a free coffee with every gym visit or 10% off athletic gear. Hosting free outdoor fitness bootcamps in parks on weekends, promoted via Instagram Reels and TikTok challenges,

would tap into their social media habits while positioning the gym as a hub for youth-centric activities.

For **Established Urban Professionals**, strategies should emphasise premium, time-efficient solutions aligned with their busy lifestyles. Integrating corporate wellness partnerships could offer subsidised gym access as part of employee benefits, coupled with lunchtime yoga or meditation classes. To maximise convenience, the gym could introduce 30-minute express workout slots with pre-booked equipment via a members-only app, alongside luxury add-ons like spa sessions or nutritionist consultations for an additional fee. Marketing efforts should target professional networks such as LinkedIn, with campaigns emphasising exclusivity like "Fitness for Career-Driven Members." Meanwhile, email newsletters could highlight time-saving features, including app-based bookings and progress tracking. Finally, curating VIP events such as wellness seminars or networking mixers would align with their desire for status and community among peers, reinforcing the gym's role as a lifestyle brand beyond basic fitness.

V. Conclusion

This report segmented 2,000 gym members using K-means++ and Agglomerative Clustering, identifying two distinct customer groups: younger, budget-conscious individuals in small towns and affluent urban professionals. After standardising age and income, cluster validity was confirmed via the Elbow Method and Silhouette Analysis (highest score: 0.54 for 2 clusters). Both methods aligned robustly, with 97% overlap, validating the segmentation's reliability. Tailored strategies, affordable programs for small-town starters, and premium services for urban affluents were proposed to enhance engagement. These insights enable the gym chain to allocate resources effectively, driving customer satisfaction and profitability.